Lsml#5

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Online learning.

X $\|X\|$. click

→ learning

>>7HD]

X

online → loss ↓↓ y_1 y_2 $w_1, w_2, w_3, \dots w_n$ → $w^* = \arg\min \sum^N L(w_i)$

f_1 f_2

$x_t \in X$

обучение

$\text{Regret}(p) =$

$\text{Regret}(p) = \sum L_i(w_i) - L_i(w^*)$

i - loss i - модель эталон

FTL = follow the leader | FTRL

$w_{t+1} = \arg\min_w \sum_i^N f_i(w_i)$

y_1 w 100%

y_2 100%

y_3 y_2 100%

y_1 y_2 y_3

f_i = loss i шаг

r_i = регул. шаг

$w_{t+1} = \arg\min_w \sum f_i(w) + r_i(w)$ $r_i = \lambda\|w - w_i\|_2^2$

→ можно в SCD

Как обучать на практике

• Strong Convexity

$f(y) \geq f(x) + \nabla f(x)^T \cdot (y - x) + m\|y - x\|_2^2$

w^* g_i кас. бу пар

Пусть loss и r_i – SC

$f \to \hat{f}$, $r \to \hat{r}$

$\hat{f}_i(w) = f_i(w_i) + \nabla f_i(w_i)(w - w_i)$ – приближение $r \to \hat{r}$ тоже

$\hat{r}_i(w) = \lambda\|w - w_i\|_2^2$

$\hat{R}egret = \sum \hat{f}_i(w_i) - \hat{f}_i(w^*)$

$Regret = \sum f_i(w_i) - f_i(w^*)$

$f_i(w_i) = \hat{f}(w_i)$

$\hat{R}egret \overset{min}{\geq} Regret \to min$

SCD

$f_i(w_i) + \nabla f_i^T(w_i)(y - y_i) + \frac{1}{2\eta}\|w - w_i\|_2^2 \to min$ $\frac{1}{2\eta} = \lambda$

$\frac{\partial}{\partial w}$

$0 + \nabla f_i(w_i) + 2\frac{1}{2\eta}(w - w_i) = 0$

$\frac{1}{\eta}(w - w_i) = -\nabla f_i(w_i)$

$w = w_i - \eta_i \nabla f_i(w_i)$ → SCD

На практике

$n \to \infty$ обобщает

$f_1 << f_0$

если обучать с L_2 → много некорр. коэф. в w ⇒ необходимо

FTRL-proximal | Info: $g_i = \nabla f_i(w_i)$, $g_{1:t} = \sum^t g_i$

$w_{t+1} = \arg\min \left(\underbrace{g_{1:t} \cdot w}_{\text{лин. пр}} + \underbrace{\frac{1}{2}\sum \sigma_s \|w - w_s\|_2^2}_{L_2} + \underbrace{\lambda\|w\|_1}_{L_1} \right)$

$\frac{1}{\eta} = \sum \sigma_s$

$= g_{1:t} \cdot w + \frac{1}{2}\sum \sigma_s \cdot (\|w\|_2^2 + \|w_s\|_2^2 - 2w^T w_s) + \lambda\|w\|_1$

$= \left(g_{1:t} - \sum \sigma_s w_s\right) \cdot w + \|w\|_2^2 \cdot \frac{1}{2\eta} + \lambda\|w\|_1 \to min$

$\boxed{g_{1:t} - \sum \sigma_s w_s} + \frac{1}{\eta_t} \cdot w + \lambda_1 \cdot \frac{2}{2\eta}\|w\|_1$

z_t

$z_t = z_{t-1} + g_t + \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}}\right) w_t$

$w = -\eta(z + \lambda\, sgn(w))$

$w < 0 \Rightarrow w = -\eta(z - \lambda)$

$w > 0 \Rightarrow w = -\eta(z + \lambda)$

$|z| \leq \lambda \Rightarrow w = 0$

$|z| > \lambda \Rightarrow w = -\eta(z - \lambda\, sgn(z))$

• SCD работает хорошо

• Если данные разр. то FRTL – лучше

$z_t + \frac{1}{\eta_t} w + \lambda \frac{\partial}{\partial w}\|w\| = 0$ $\to +1, -1$

$z + \frac{1}{\eta} w + \lambda\, sgn(w) = 0$

$f_i(w_i) + \nabla f_i(w_i)(w^* - w_i) + \lambda\|w^* - w_i\|_2^2$ $\geq$ =

$\wedge\!\!\vee$

$f(w^*)$